Research Proposal

Evidence-Gated Medical LLM Alerts for Pharmacogenomic Claims

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Thesis: The capstone is now deliberately pruned. It studies one clinically interpretable workflow: LLM-drafted pharmacogenomic or genomic medication-alert text. The evaluation asks whether a pre-display evidence gate reduces overclaiming relative to guideline-supported evidence without inappropriately denying bounded, aligned alerts.

Problem Statement

Medical LLMs can sound authoritative even when the evidence only supports a narrower statement. In pharmacogenomics this matters because a real guideline, a verified citation, a population-frequency result, or a variant assertion may support a medication-safety note without proving broad patient-outcome benefit, universal population transport, or deterministic action.

Hypothesis

Compared with ungated LLM alert text, a three-gate controller should reduce evidence overclaiming while tracking inappropriate denial in a future independent evaluation. The supported Stage 1 claim is narrower: the implementation follows its intended evidence grammar on author-designed synthetic archetypes.

Methods

- Workflow: pharmacogenomic/genomic medication-alert text, including medication-gene pairs and variant assertions relevant to prescribing or safety review.
- Gates: endpoint/actionability, population fit, and citation/guideline support.
- Stage 1: 30 author-designed synthetic cases with expected decisions; executable controller reports specification conformance, designed overclaim archetypes blocked, bounded alerts allowed, inappropriate denial, and error categories.
- Stage 2: replace author-designed cases with independently authored cases and blinded reviewer adjudication by qualified clinicians or pharmacogenomics reviewers.

Stage 1 Progress

Metric	Current result	Interpretation
Cases	30	Synthetic case bank for the pruned workflow.
Author-designed overclaim share	20/30	Stress-test coverage, not a measured clinical base rate.
Remaining after gate	0/30	Specification-conformance check, not clinical safety.
Overclaim archetypes blocked / bounded alerts allowed	20/20; 10/10	Conformance counts, not independent accuracy.
Inappropriate denial	0	Count of bounded aligned alerts blocked too strongly.

Value

The contribution is a submission-ready narrowing move: instead of presenting a universal assurance theory, the paper proposes a small auditable benchmark for a recurrent clinical workflow. This directly answers the critique that the prior package was broad without a single causal chain.

Selected Sources

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